IPA ASSIGNMENT - 24

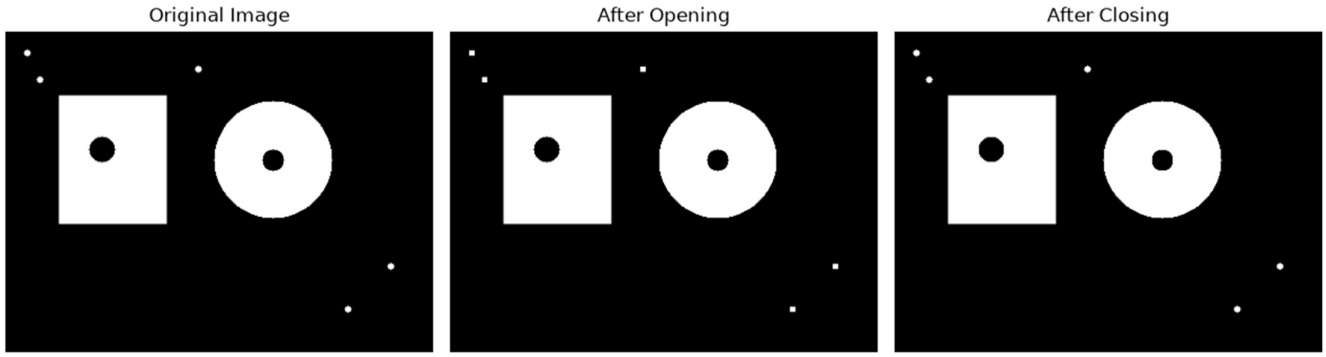
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1. Opening and Closing Detection

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("img_1.jpeg")
if img is None: raise FileNotFoundError("Image not found! Check the
filename.")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))
opening = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)
closing = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel)
plt.figure(figsize=(12, 5)) plt.subplot(1, 3, 1)
plt.imshow(binary, cmap="gray") plt.title("Original Binary Image")
plt.axis("off") plt.subplot(1, 3, 2)
plt.imshow(opening, cmap="gray") plt.title("Opening")
plt.axis("off") plt.subplot(1, 3, 3)
plt.imshow(closing, cmap="gray") plt.title("Closing")
plt.axis("off") plt.tight_layout() plt.show()
```

OUTPUT



2. Dilation Image

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread("img_2.jpeg")
```

```
if image is None:raise FileNotFoundError("Image not found! Check the
filename.")
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
_, binary_image = cv2.threshold(gray_image, 127, 255, cv2.THRESH_BINARY)
kernel = np.ones((7, 7), dtype=np.uint8)
dilated_image = cv2.dilate(binary_image, kernel, iterations=1)
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1) plt.imshow(binary_image, cmap="gray")
plt.title("Original Binary Image") plt.axis("off")
plt.subplot(1, 2, 2) plt.imshow(dilated_image, cmap="gray")
plt.title("Image After Dilation") plt.axis("off")
plt.tight_layout() plt.show()
```

OUTPUT

